



VBF-T7R3-DFMEA-01 Case t7_r3 | 2026-08-26

Prepared: _____ Reviewed: _____ Approved: _____

Virtual design package - simulation caliber (no physical build)

VBF Design FMEA (qualitative version, based on simulation signals)

****Document number**:** VBF-T7R3-DFMEA-01 · ****Case**:** t7_r3 · ****Date**:** 2026-08-26

Qualitative S/O ratings are derived from the magnitude of simulation values vs thresholds. Complete FMEA including process/supplier failures: N/A (beyond pure simulation boundary). Prepared / Reviewed / Approved:

#	Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	SxO risk	Design-side mitigation
1	Negative electrode plating during 4C charge	electrolyte concentration polarization / anode overpotential	anode_potential_v min +0.0171 V (margin only 17 mV)	High	**Low**	(full CC charge shows 0 negative instants)	Medium t+ 0.6 + D■ 1×10 ⁻⁹ m ² /s + 2.0 μm anode particles implemented; margin watch: any electrolyte degradation (t+ drift) in service re-opens risk -> consider N/P 1.05-1.15 for production intent
2	Thermal runaway by nail penetration	internal short + heat accumulation > cooling capacity	triggered=false only with hA >= 0.3 W/K; at 0.2 W/K T=405 K at 2097 s (near trigger)	High	**Low**	(designed cooling 0.5 W/K covers 1.7× margin)	Medium mandatory active cooling specification hA = 0.5 W/K; add BMS redundancy + cooling-loss derating; fail-safe on cooling failure
3	Thermal runaway by 4C-charge heat-up	4C charge energy dissipation in lumped cell	T_max 363.2 K (90 °C) during 4C - elevated but stable (no TR criterion)	Medium	Medium	Medium	charge de-rate above 45 °C ambient; monitor T in BMS; the 363 K value is a virtual lumped estimate
4	Electrolyte oxidative decomposition at high voltage	HOMO vs cathode potential mismatch	**No signal available**	molecular stage skipped (`real_compute=false`, start_stage 3 - HOMO/IE-EA not computed).	NMC811 @4.2 V known-lower risk class (literature)	High	Unknown (unassessed) High (unassessed) Stage-2 true compute (DFT HOMO/IE-EA) recommended as follow-up; keep cut-off <= 4.2 V
5	SEI over-thickening (capacity/ED fade)	SEI growth per ec-reaction-limited model	SEI end 512.9 nm vs 550 nm threshold (6.7 % margin after 100 cyc @45 °C)	Medium	Medium	Medium	FEC additive proposed (unvalidated in this run); lower-temperature operation; the metric is a virtual SEI-thickness proxy, not a measured capacity-loss curve
6	Insufficient capacity / energy density	architecture trade-off	ED 482.4 vs 327.18 Wh/kg (47 % margin); 1C cap 6.257 Ah	Low	Low	Low	stability confirmed
7	N/P imbalance (0.862 < industry 1.05-1.15)	negative thinner than industry practice	derived areal balance 0.862; plating still passes at 4C	Medium	Low (virtual)	Medium	production intent: raise N/P -> thicker anode (+ plating margin via more anode capacity, -ED ~5 %)

****Conclusion**:** Highest-risk items are #4 (electrolyte oxidation - unassessed because molecular stage skipped under `real\_compute=false`) and #2/#3 (thermal-knife-edge - mitigated by the mandatory 0.5 W/K cooling design). All mitigations that live inside the virtual cell design have been implemented; those needing physical/material follow-up (#4 Stage-2 compute, production N/P, physical TR tests) are listed for the release book.